ECE 2305 – Introduction to C Programming
Tutorial 10
Logic Operations

Write a C++ program to demonstrate the Boolean operations by creating the Truth Tables

```
bool A;
bool B;
```

NOT `(!A)`
Shown the result for `A = 0` and `A = 1`.

AND `(A && B)`
OR `(A || B)`
NAND `!(A && B)`
NOR `!(A || B)`
XOR `(A && !B || !A && B)`
XNOR `(A && B || !A && !B)`
Shown the result for

```
A = 0 , B = 0.
A = 0 , B = 1.
A = 1 , B = 0.
A = 1 , B = 1.
```

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Input

```
// ECE 2305 - Tutorial 10
// Logic Functions
#include<iostream>
using namespace std;

int main()
{
	bool A;
	bool B;
	cout << "Boolean Operations" << endl;
	cout << endl;

	cout << "A \t(!A)" << endl;
	A = 0, cout << A << " " << "\t" << (!A) << endl;
	A = 1, cout << A << " " << "\t" << (!A) << endl;
	cout << endl;

	cout << "A B \t(A && B)" << endl;
	A = 0, B = 0, cout << A << " " << B << "\t" << (A && B) << endl;
	A = 0, B = 1, cout << A << " " << B << "\t" << (A && B) << endl;
	A = 1, B = 0, cout << A << " " << B << "\t" << (A && B) << endl;
	A = 1, B = 1, cout << A << " " << B << "\t" << (A && B) << endl;
	cout << endl;

	cout << "A B \t(A || B)" << endl;
	A = 0, B = 0, cout << A << " " << B << "\t" << (A || B) << endl;
	A = 0, B = 1, cout << A << " " << B << "\t" << (A || B) << endl;
	A = 1, B = 0, cout << A << " " << B << "\t" << (A || B) << endl;
	A = 1, B = 1, cout << A << " " << B << "\t" << (A || B) << endl;
	cout << endl;

	cout << "A B \t!(A && B) = (!A || !B)" << endl;
	A = 0, B = 0, cout << A << " " << B << "\t" << !(A && B) << endl;
	A = 0, B = 1, cout << A << " " << B << "\t" << !(A && B) << endl;
	A = 1, B = 0, cout << A << " " << B << "\t" << !(A && B) << endl;
	A = 1, B = 1, cout << A << " " << B << "\t" << !(A && B) << endl;
	cout << endl;

	cout << "A B \t(A || B)" << endl;
	A = 0, B = 0, cout << A << " " << B << "\t" << !(A || B) << endl;
	A = 0, B = 1, cout << A << " " << B << "\t" << !(A || B) << endl;
	A = 1, B = 0, cout << A << " " << B << "\t" << !(A || B) << endl;
	A = 1, B = 1, cout << A << " " << B << "\t" << !(A || B) << endl;
	cout << endl;

	cout << "A B \t(A && !B || !A && B)" << endl;
	A = 0, B = 0, cout << A << " " << B << "\t" << (A && !B || !A && B) <<
endl;
	A = 0, B = 1, cout << A << " " << B << "\t" << (A && !B || !A && B) <<
endl;
	A = 1, B = 0, cout << A << " " << B << "\t" << (A && !B || !A && B) <<
endl;
	A = 1, B = 1, cout << A << " " << B << "\t" << (A && !B || !A && B) <<
endl;
	cout << endl;
```

```
	cout << "A B \t(A && B || !A && !B)" << endl;
	A = 0, B = 0, cout << A << " " << B << "\t" << (A && B || !A && !B) <<
endl;
	A = 0, B = 1, cout << A << " " << B << "\t" << (A && B || !A && !B) <<
endl;
	A = 1, B = 0, cout << A << " " << B << "\t" << (A && B || !A && !B) <<
endl;
	A = 1, B = 1, cout << A << " " << B << "\t" << (A && B || !A && !B) <<
endl;
	cout << endl;

	system("pause");
	return(0);
}
```

Output

